观一间铺，算一门生意。
Find the opportunity. Measure the risk.

Front Matter



An AI-powered storefront feasibility platform that helps small merchants evaluate space, rent, risk, and business potential before signing a lease.

观铺是一款 AI 商铺可行性评估平台，帮助小商户在签约前，通过店铺照片和租赁信息判断空间是否适合、租金是否合理、风险是否可控。

http://leaselens.ycxdacongming.icu/

风险从未如此昂贵

Lease Risk Is Rising.

Singapore's F&B market still has demand, but closures show how unforgiving lease decisions have become. For **small merchants**, a wrong lease can turn rent, renovation, and ramp-up time into fixed pressure before the business reaches stable revenue.

307 / month

F&B Closures

Singapore averaged 307 F&B closures per month in 2025, up from 254 per month in 2024.

Closed Within 5 Years

Over 60% of F&B outlets that closed from January to October 2025 had operated for less than five years.

60%

82%

Never Profitable

Among closed F&B outlets under five years old, 82% had never made a profit.

⚠ **The lease is the first stress test of the business.** ⚠



痛点 THE SHOPFRONT PROBLEM

铺面不只是空间，而是一笔每个月都必须兑现的风险。

A Storefront Is Not Just A Space. It Is A Monthly Bet.

A founder sees an empty unit and imagines a café, studio, or boutique. But before the first customer walks in, the lease has already started counting: rent, renovation, utilities, staffing, and months of cash-flow pressure.



Can the RENT be survived?

Fixed rent starts before stable revenue does. A promising concept can fail if the monthly burn is too heavy from day one.

Does the SPACE fit the business, not just the photo?

Listing photos show surfaces. Real operations need layout, utilities, ventilation, storage, service flow, and back-of-house logic.

Will customers NOTICE it?

Foot traffic only matters when the storefront can convert attention into visits. Visibility, frontage, signage, and sightlines decide whether people walk in.

When does the lease start PAYING BACK?

A good idea still fails if rent, renovation, ramp-up time, ticket size, and daily traffic do not add up to a realistic payback period.

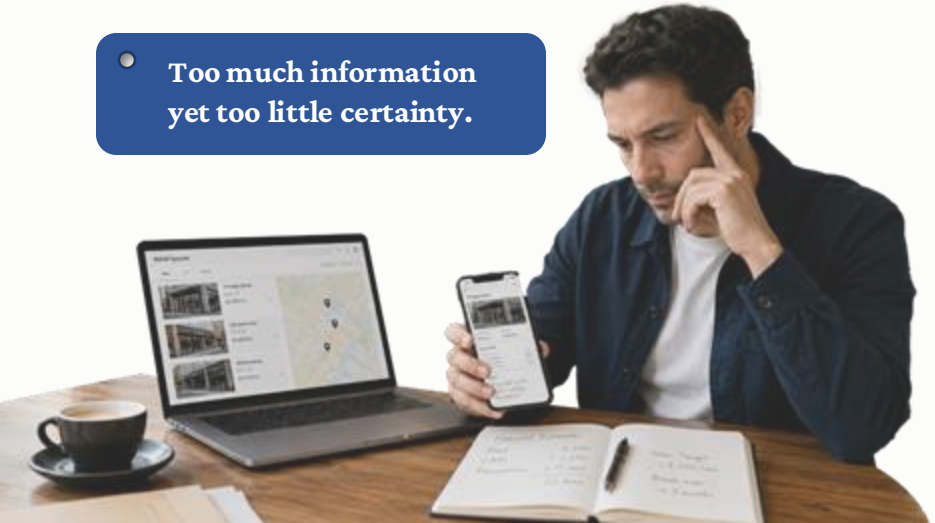
现状 CURRENT METHOD

他们不是没有尽调，而是证据没有连成判断。

Founders do the work. The evidence stays fragmented.

Before signing, small founders usually try to check the site. They ask brokers, compare listing photos, search the area on Google Maps, visit the unit, and build a rough spreadsheet. But these inputs remain disconnected. They do not combine rent, space, competition, infrastructure, and payback into one clear decision.

Too much information yet too little certainty.



BROKER ADVICE 01.

Helpful for finding units, but not neutral enough to judge whether the tenant's business can survive the lease.

LISTING PHOTOS 02.

Show what the space looks like, but hide the operational details: utilities, ventilation, signage, and fit-out difficulty.

GOOGLE MAPS 03.

Shows what is nearby, but not whether the location can convert traffic into sustainable revenue.

ROUGH SPREADSHEET 04.

Adds rent and cost assumptions, but rarely tests downside scenarios, payback, or rent pressure together.

MISSING ANSWER 05.

After all the checking, one question remains: can this specific lease actually work? Can my business actually survive?

同样是看铺，大品牌看的是模型，小商户赌的是判断。

Same storefront. Different visibility.

• Data Sources & Analytical Capabilities

❖ Large Chains & Developers

- ✓ In-house property teams & consultants
- ✓ Location intelligence & footfall analytics
- ✓ Competitor density & catchment modelling
- ✓ Scenario-based financial modelling

❖ Independent Merchants & Café Founders

- ✗ Broker recommendations (conflicted)
- ✗ Google Maps & listing photos
- ✗ Informal word-of-mouth advice
- ✗ Personal spreadsheets, often incomplete

Structured data + rigorous modelling



Fragmented inputs + gut feel

同样承担租约风险，小商户却缺少判断风险的系统。

The same lease risk, but not the same decision tools.

• Execution & Risk Management

❖ Large Chains & Developers

- ✓ Lease negotiation specialists
- ✓ Multi-site benchmarking before commitment
- ✓ In-house legal teams
- ✓ Signed leases with modelled risk

Controlled execution + risk transparency

❖ Independent Merchants & Café Founders

- ✗ Same conflicted brokers
- ✗ No structured benchmarking
- ✗ No legal leverage
- ✗ Signed leases with unmodelled risk

Reactive decisions + hidden risk

市场仍有动能

The Market Demand Is Still There.



Singapore retail & F&B remain active:

4.8%
Retail Sales Growth
Retail trade sales increased year-on-year in March 2026

2.3%
F&B Sales Growth
F&B services sales increased year-on-year in March 2026

16.9M
Visitor Arrivals
Singapore welcomed 16.9 mil international visitors in 2025

S$32.8B
Tourism Receipts
Tourism receipts reached S$32.8 billion in 2025

But a strong market doesn't make every site viable.
Success starts with **the right location.**

Can this site turn rent, space, and business model into a profitable store?

解法 THE SOLUTION

把看不见的租约风险，变成可判断的商业报告。

FrontMatter makes lease decisions visible.

Upload a storefront or interior photo. Enter the rent, area, renovation budget, and target business type. FrontMatter analyzes the opportunity through spatial fit, rent pressure, business suitability, and hidden lease risks before the user signs.

1 Input

Shop photos + lease details

Shop photos + lease details: rent, area, budget, business type

2 AI Evaluation

Space, finance, and risk analysis

FrontMatter reads spatial signals, estimates rent pressure, checks business fit, and flags operational risks that photos alone cannot reveal.

3 Business Case

A decision-ready feasibility report

The user receives scores, warnings, recommendations, and next-step actions before committing to the lease.



流程 WORKFLOW

One workflow connects image, lease data, location context, and financial risk.

RAW INPUTS

- Photo
- Lease terms
- Location

01 Upload Site Image

Upload PNG / JPG / WEBP storefront or interior photos, up to 10 MB.

02 Enter Lease Profile

Input monthly rent, floor area, lease term, service charge, fit-out budget, business type, and operating assumptions.

03 Complete F&B Readiness

Confirm cooking intensity, floor level, layout, water, power, gas, floor trap, grease trap, exhaust, wastewater, and approved use.

04 Resolve Location

Choose "At site now" or "Search address". Address autocomplete and resolution run through server-side Google Places, keeping the API key off the frontend.

05 Map Market Signals

The system **observes** nearby verified merchants, proximity levels, and commercial surroundings. These are treated as market context, not guaranteed footfall proof.

06 Run Scoring + Cash Flow

Applies rule-driven scoring and lease-term modelling: 0–100 score, NPV, IRR, discounted payback, break-even daily customers, monthly cash flow, BASE / DOWN -10% / SEVERE -25% scenarios.

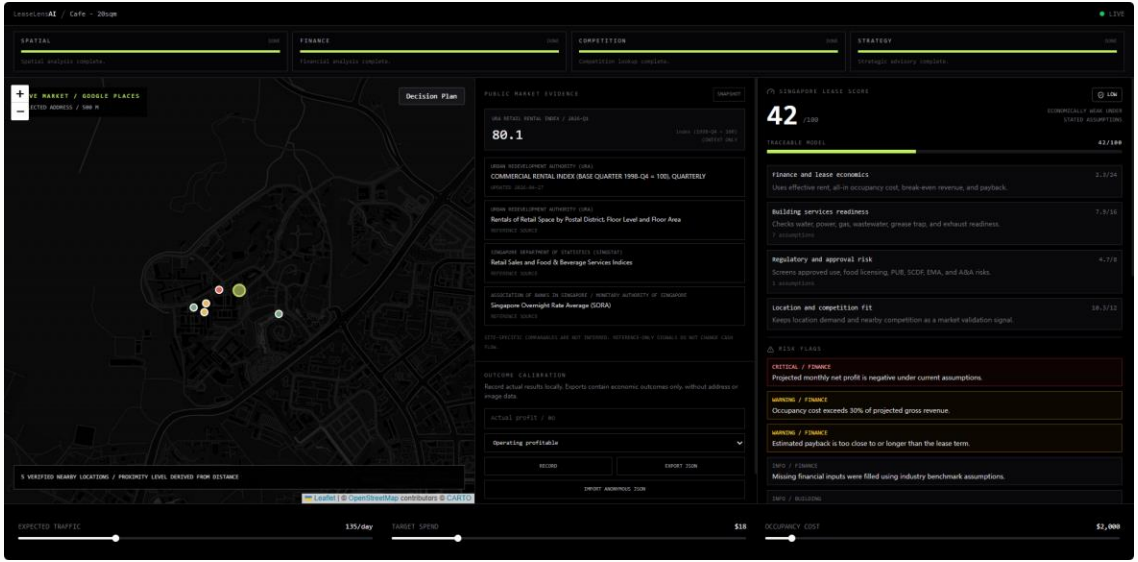
07 Decision Report

The result page **returns** a score breakdown, risk flags, confidence level, verdict, spatial blueprint, market observations, candidate-site comparison, and What-if controls for traffic, ticket size, and rent.

A lease decision is turned into a structured report.

Input: Input: Café · 20 sqm · Singapore storefront · rent, traffic, target spend, F&B readiness

Current assumptions show weak lease viability. The site needs either lower occupancy cost, stronger expected traffic, or higher target spend before it becomes financially defensible.



Lease Score: 42 / 100

Economically weak under stated assumptions. The score is traceable across finance, building readiness, regulatory risk, and location fit.

Key Risk Flags

Projected monthly net profit is negative. Occupancy cost exceeds safe revenue range. Estimated payback is too close to, or longer than, the lease term.

Decision Workspace

The report combines market evidence, map context, score breakdown, and what-if controls so users can test the lease before signing.

核心功能 CORE FEATURES

让租赁风险可见、可测、可比较。

Built to make lease risk Visible, Testable, and Comparable.

FrontMatter is a locally deployable AI system for evaluating commercial leases in Singapore. Users upload a site image and enter rent, area, lease term, fit-out budget, business type, and F&B readiness details. The system produces a viability score, risk explanation, operational blueprint, and recommended candidate locations.

观铺 FrontMatter 不是普通图片识别或聊天机器人。它结合 AI 视觉、确定性评分、位置数据和财务模拟，把一个铺面转化成结构化的租赁判断。每个核心功能都服务于同一个问题：这个店址到底能不能成为一门可持续的生意？



功能一 HYBRID SCORING

FEATURE 1

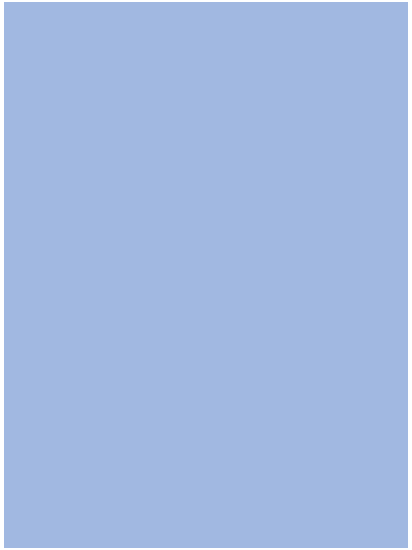
HYBRID SCORING

混合评分

Hybrid Scoring

60% rules
40% AI judgement.

Deterministic rules evaluate rent pressure, infrastructure, regulatory use risk, and local market conditions. AI judgement adds image-based spatial reading and expert-style reasoning.



FEATURE 2

F&B READINESS CHECK

FEATURE 3

MARKET SIGNAL LAYER

FEATURE 4

WHAT-IF SIMULATE

功能二 F&B READINESS CHECK

FEATURE 1
HYBRID SCORING

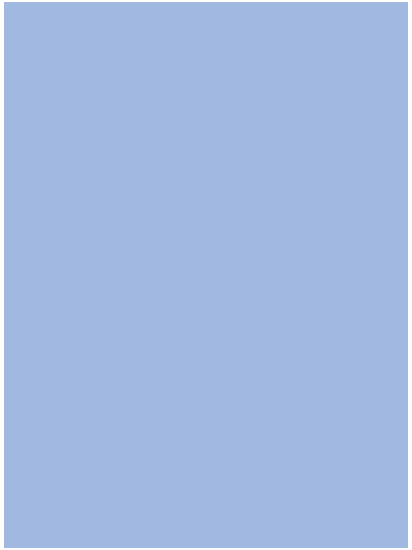
FEATURE 2
F&B READINESS CHECK

餐饮开店尽调

F&B Readiness Check

Looks suitable is not the same as ready to operate.

The system checks water, power, gas, floor trap, grease trap, exhaust, wastewater, and approved use before the user commits to a café, bakery, restaurant, or bar concept.



FEATURE 3
MARKET SIGNAL LAYER

FEATURE 4
WHAT-IF SIMULATE

功能三 MARKET SIGNAL LAYER

FEATURE 1
HYBRID SCORING

FEATURE 2
F&B READINESS CHECK

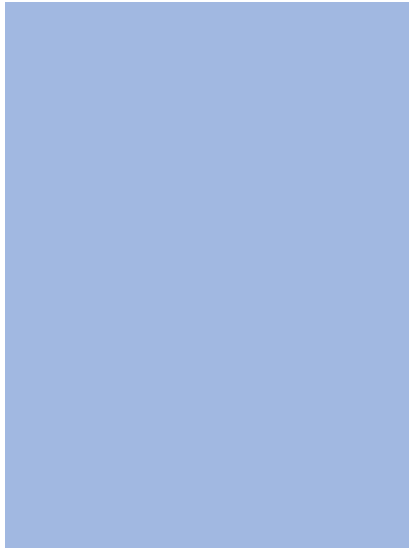
FEATURE 3
MARKET SIGNAL LAYER

周边信号分析

Market Signal Layer

A storefront is judged with its surroundings.

Google Places data helps FrontMatter read nearby competitors, commercial density, surrounding demand signals, and candidate alternative locations.



FEATURE 4
WHAT-IF SIMULATE

功能四 WHAT-IF SIMULATION

FEATURE 1
HYBRID SCORING

FEATURE 2
F&B READINESS CHECK

FEATURE 3
MARKET SIGNAL LAYER

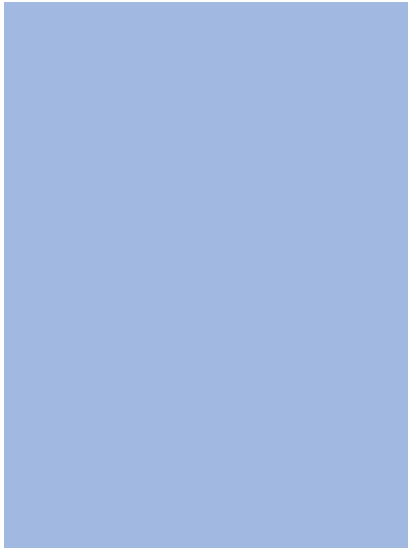
FEATURE 4
WHAT-IF SIMULATE

经营假设模拟

What-if Simulation

Stress-test the lease before signing.

Users can adjust rent, footfall, ticket size, gross margin, and renovation budget to see how each assumption changes payback period, rent pressure, and overall viability.



Frontend Intake

01. Intake Schema

Next.js · React · TypeScript · Tailwind · Leaflet power the web UI for image upload, lease forms, F&B inputs, maps, and what-if controls

02. Vision Processing

GLM **glm-4.1v-thinking-flash** converts PNG / JPG / WEBP site images into a structured **spatialBlueprint**.

03. Location Grounding

Server-side Google Places handles autocomplete, address resolution, nearby merchants, and **proximityLevel** without exposing API keys.

Backend Orchestration

04. Backend Orchestration

FastAPI · SQLAlchemy · AsyncPG · Alembic · Redis validate inputs, coordinate analysis, and serve auth, location, report, and calibration APIs.

05. Rule-Based Scoring

Structured rules generate the 0–100 score, **scoreBreakdown**, **riskFlags**, confidence level, and verdict while the LLM supports interpretation.

06. Financial Engine

The DCF model calculates **NPV, IRR, discounted payback, break-even daily customers,** monthly cash flow, and BASE / DOWN / SEVERE scenarios.

07. Decision Workspace

The result page combines **report**, map observations, public evidence, candidate-site comparison, and live traffic / ticket / rent what-if controls.

08. Decision Workspace

PostgreSQL + pgvector · Redis · Nginx · Docker Compose support storage, runtime services, public routing, and containerized deployment.

核心优势 WHY IT WINS

看得见铺面，也算得清生意。

Generic AI describes. Portals list. Front Matter decides.

Traceable score. Official data. Risk screening. Financial model. All live.

CORE METRIC	GENERIC AI	PROPERTY PORTALS	FRONT MATTER (LIVE)
Spatial Analysis	Visual description	No image processing	**F&B Suitability Audit & Scoring**
Financial judgement	Nowhere to start calculation	Displays static rent	**Breakeven & Payback Model**
Compliance Risk	None	None	**PUB / SCDF / EMA Screen**
Data source	Outdated global web	Proprietary listings	**URA & SINGSTAT API**
Simulation	None	None	**Real-time "What-If" Engine Adjust assumptions → see impact**
Output Action	Scattered prompts	"Contact Agent" form	**Targeted Operating Plan**

团队 CROSS-DISCIPLINARY TEAM

Engineering builds the system. Finance defines the case. Data measures the risk. Product makes it clear.

Ye Chixia
叶驰遐



Technical Lead / System Architect
MSc in Computer Control & Automation, Nanyang Technological University

"Build the engine that makes the decision possible."

Gao Qi
高琪



Business & Financial Modeling Lead
Digital Financial Technology, NUS School of Computing, Undergraduate Financial Management

"Turn lease terms into business logic."

Yu Hanyue
俞涵悦



Analysis & Product Strategy Lead
Undergraduate of Mathematics and Data Science Analytics, National University of Singapore.

"Translate market insights into strategic clarity."

Liu Ziyi
刘子羿



Product & Spatial Intelligence Lead
Undergraduate of Architecture, National University of Singapore.

"Translate space into understandable decisions."

市场机会 MARKET OPPORTUNITIES

现实差距

The Reality Gap.

MACRO DEMAND

+4.8% Retail YoY Growth

+2.3% F&B Sales Growth

LEASE BLINDSPOT

307 F&B Closures / Month

60% Outlets Close < 5 Yrs

TARGETED AUDIENCE & WILLINGNESS TO PAY

Segment	Core Pain Point	Willingness to Pay
P0 First-time Founders	High risk aversion; afraid of regulatory/lease traps	**Highest** (Avoids $150k bankruptcy)
P1 Multi-store Brands	High-frequency site benchmarking & cross-comparison	**Medium-High** (SaaS subscription)
P2 Property Brokers	Needs data-backed reports to close deals faster	**B2B Volume** (Annual contract license)

以新加坡作为可复制的首个切入市场

SINGAPORE AS A PORTABLE BEACHHEAD

Extreme Density & High Turnover

高密度和高流转

One of the highest F&B densities globally. High lease friction triggers persistent, recurring data requests.

Transparent Regulatory Framework

监管环境清晰透明

Standardized URA codes, PUB drainage rules, and SCDF codes make deterministic rule-modeling highly effective.

Global Launchpad Value

全球跳板价值

The English-based, standardized legal model is easily portable to high-rent global hubs: **Hong Kong → Dubai → London.**

商业模式 BUSINESS MODEL

From a single lease decision to **commercial site selection infrastructure.**

PART 1: TO CUSTOMER (TOC)

Ecosystem Bootstrapping

FREE

- **Frictionless Onboarding:**
  Helps individual F&B entrepreneurs effortlessly avoid legal and leasing traps in Singapore.
- **Freemium Index Preview:**
  Delivers baseline spatial data and partial readiness scores to trigger mass property inquiries.
- **Data Flywheel:**
  Leverages high-frequency, user-uploaded shop photos and site locations to train our spatial AI engine at zero cost.

PART 2: TO BUSINESS (TOB)

Enterprise Solutions & On-Premise

Custom Pricing

- **API Integration (Solution):**
  Seamlessly injects our proprietary F&B hardware audit engine (URA, PUB, SCDF) into local property listing websites and apps as an authoritative "F&B-Ready Badge."
- **On-Premise Deployment:**
  Provides isolated, enterprise-grade site selection pipeline software for expanding F&B chains to safeguard strategic commercial data.
- **High-LTV Monetization:**
  Bypasses frontend noise to directly capture infrastructure-level utility budgets from B2B platform partners.

Flywheel: ToC generates the data walls; ToB integrations extract the core financial value.

落地 DEPLOYMENT & SCALE-UP

01. Now — Finish Product

Live-deployed prototype

- A public demo is running with login access.

End-to-end workflow

- Image upload, lease inputs, F&B readiness, Google Places location, score, risk flags, cash-flow model, and decision report.

Technical stack working

- Next.js + FastAPI + PostgreSQL/pgvector + Redis + Nginx + Docker Compose.

02. Next Steps — Technical

Data grounding

- Integrate OneMap, URA rental references, and more Singapore-specific geospatial/market datasets.

Scoring calibration

- Compare predictions with real outcomes to refine weights and risk thresholds..

Report export

- Generate clean PDF / investor-ready reports for founders, partners, landlords, SME advisors.

Production Hardening

- Add user accounts, permissions, audit logs, secure storage, and retention rules.

03. Next Steps — Business

Founder testing

- Run pilot tests with café founders, small F&B operators, and first-time tenants.

Pricing validation

- Test whether users pay per report before signing a lease.

B2B partnerships

- Approach brokers, incubators, SME advisors, and landlords who need structured pre-lease screening.

Vertical expansion

- Start with Singapore F&B, then expand to retail, studios, clinics, and service businesses.

发展愿景 VISION & ROADMAP

From working prototype to the unified global commercial lease intelligence platform, democratizing Location Intelligence on a Global Scale.

PHASE 01 Q2 – Q4 2026

Grounding & Security

➢ **Geospatial Integration | 官方数据打底**
Integrate Singapore OneMap, URA & SingStat APIs for exact structural demographics.

➢ **Rental Benchmarking | 租金基准核验**
Establish localized active rent baselines and operating cashflow templates.

➢ **Enterprise Security | 安全审计机制**
Deploy production-grade RBAC controls, secure auth, and pre-lease audit logs.

PHASE 02 Q1 – Q2 2027

Flywheel & Expansion

➢ **Data Flywheel | 用户贡献飞轮**
Leverage user-submitted leases and storefront photos to train the spatial AI at zero cost.

➢ **Multi-Sector Expansion | 多业态拓展**
Scale pre-lease feasibility engines from F&B to retail, wellness, and clinics.

➢ **One-Click Report | 决策报告一键导出**
Generate exportable, investor-ready feasibility PDFs for small business founders.

PHASE 03 2027 & beyond

Standard & Portability

✓ **FrontMatter Score | 行业公信标**
Turn the proprietary risk score into a trusted benchmark for tenants and brokers.

✓ **Accuracy Calibration | 闭环数据校准**
Anonymize actual store performance data to calibrate and refine predictive NPV/IRR.

✓ **Global Scaling | 跨国沙盒复制**
Port Singapore's structured model to high-rent hubs: Hong Kong, London, and Dubai.